

2.2 Probability axioms

We model random phenomena using the following formalism:

Definition 2.2. A **probability space** is a triple $(S, \mathcal{A} \subseteq \mathcal{P}(S), P)$:

- **(sample space)** A nonempty set S containing all of the possible outcomes of the phenomenon;
- **(events)** Subsets $A \subseteq S$ of the sample space – sets of possible outcomes;
- **(probability measure)** A function P that takes an event A and assigns to it a real number called $P(A)$. This number is the probability that the ultimate outcome of the random phenomenon is an element of the subset A .

In order for the probability measure P to be valid, we require that it satisfy the following *axioms*:

Definition 2.3. The **axioms of probability** are:

1. **(total measure 1)** $P(S) = 1$ (“something in the sample space is guaranteed to happen”);
2. **(nonnegativity)** $P(A) \geq 0$ for any event $A \subseteq S$;
3. **(countable additivity)** If $A_1, A_2, A_3, \dots$ is a countably infinite sequence of pairwise disjoint events (meaning $A_i \cap A_j = \emptyset$ for any distinct i, j), then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

The remainder of this course is in some sense an extended investigation of the downstream consequences of those three innocent-looking axioms: total measure 1, nonnegativity, and countable additivity. With these raw materials, we will build a whole cathedral that is necessary for successful human reasoning, useful for science and technology, and most importantly, beautiful to behold!

Notational Point 2.1. The infinite union notation we just introduced simply means

$$\bigcup_{i=1}^{\infty} A_i = A_1 \cup A_2 \cup A_3 \cup \dots$$

It is meant in the exact same spirit as if we had an infinite sequence of numbers $a_1, a_2, a_3, \dots$, and we wrote

$$\sum_{i=1}^{\infty} a_i = a_1 + a_2 + a_3 + \dots$$

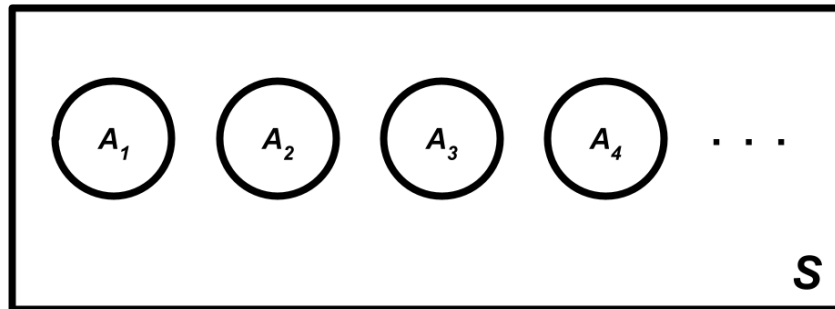


Figure 2.1

We will not prove it, but you are free to use the fact that all of our algebraic properties from Section 1.6 extend to the infinite case. So for instance,

$$\begin{aligned}
 B \cap \bigcup_{i=1}^{\infty} A_i &= B \cap (A_1 \cup A_2 \cup A_3 \cup \dots) \\
 &= (B \cap A_1) \cup (B \cap A_2) \cup (B \cap A_3) \cup \dots \\
 &= \bigcup_{i=1}^{\infty} (B \cap A_i) \quad \text{(distributive property)}
 \end{aligned}$$

$$\left(\bigcap_{i=1}^{\infty} A_i \right)^c = \bigcup_{i=1}^{\infty} A_i^c \quad \text{(De Morgan)}$$

Feel free to use these things.

Remark 2.2. The axiom of countable additivity is the most practically useful of the three, because that is the axiom that opens the door to actually *calculating something*. It provides a condition under which the problem of “computing a probability” collapses to a problem of “adding.”

Remark 2.3. The axiom of countable additivity also gives us our first warning that calculus will be necessary for studying probability. Apparently it is possible that a probability might take the form of an *infinite series*, and so we may have to simplify those.

Remark 2.4. Figure 2.1 displays a cartoon of a countably infinite sequence of pairwise disjoint sets. So, an infinite parade of blobs in the plane that do not overlap at all. The axiom of countable additivity says that the total probability of all of these sets taken together (their union) is equal to the sum of the individual probabilities. You should note that this is exactly how area works as well. If you wanted the total area covered by all of those blobs, you would just add them up. This is intuition that you can take to the bank. Probability behaves exactly the same way that length, area,

and volume behave on the number line, the plane, and three-dimensional space, respectively. This is because these are all special cases of the same abstract mathematical structure. So anything you know about how length/area/volume behave can probably be applied one-for-one with probability, and you will get the right answer.

Remark 2.5. In some sense, probability theory is “just” a branch of mathematical analysis. In particular, it is a branch of measure theory, and probability spaces are special cases of measure spaces (as are Euclidean spaces with their measures of length, area, volume, etc). If you are interested in really going deep on the mathematics of probability (something you will have to do if you are considering graduate work in statistics), I suggest you take courses in real analysis.

To close this section, let’s dip our toes in and use the axioms to prove two simple but useful results that we will need very shortly

Theorem 2.1. $P(\emptyset) = 0$.

Proof. Consider the countably infinite sequence of sets $A_1 = \emptyset$, $A_2 = \emptyset$, $A_3 = \emptyset$, and so on. So $A_i = \emptyset$ for all $i \in \mathbb{N}$. Since $\emptyset \cap \emptyset = \emptyset$, this sequence is pairwise disjoint. Furthermore,

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} \emptyset = \emptyset.$$

As such, the axiom of countable additivity guarantees that

$$P(\emptyset) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) = \sum_{i=1}^{\infty} P(\emptyset) = \lim_{n \rightarrow \infty} \sum_{i=1}^n P(\emptyset) = \lim_{n \rightarrow \infty} n \cdot P(\emptyset).$$

The only way for that limit on the right to converge to a finite number is if $P(\emptyset) = 0$. □

Riveting stuff, but you forget it at your peril. And in fact, we *need* this result to prove something that may seem obvious, but we haven’t actually established yet. The axiom of countable additivity tells us how to compute the probability of a (countably) infinite union of disjoint events. To do it, you evaluate an infinite sum. What if I want the probability of a *finite* union of disjoint sets? Is that a finite sum? Mercifully, yes. But why? It may seem intuitively “obvious” to you that if we have it in the infinite case we *must* have it in the simpler finite case, but at this point we have no formal justification for that. It’s just vibes. So let’s correct that:

Corollary 2.2. (finite additivity) If $A_1, A_2, \dots, A_n$ is a finite collection of pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i).$$

Proof. We already have $A_1, A_2, \dots, A_n$, so let us now artificially extend this collection to an infinite sequence where $A_j = \emptyset$ for all $j > n$. We *need* to augment the finite collection into an infinite collection because infinite collections are the only thing we know how to compute with at this stage. But conveniently, it turns out that the union of our shiny new infinite collection is in fact exactly equal to the union of the original finite collection:

$$\begin{aligned}
\bigcup_{j=1}^{\infty} A_j &= \left(\bigcup_{j=1}^n A_j \right) \cup \left(\bigcup_{j=n+1}^{\infty} A_j \right) \\
&= \left(\bigcup_{j=1}^n A_j \right) \cup \left(\bigcup_{j=n+1}^{\infty} \emptyset \right) \\
&= \left(\bigcup_{j=1}^n A_j \right) \cup \emptyset \\
&= \bigcup_{j=1}^n A_j.
\end{aligned}$$

So we've turned the finite collection into an infinite one by padding out the rest of the sequence with $\emptyset$, but we haven't actually changed the contents of anything. Now, applying the axiom of countable additivity exactly as stated, we can calculate the probability of the finite union:

$$\begin{aligned}
P\left(\bigcup_{j=1}^n A_j\right) &= P\left(\bigcup_{j=1}^{\infty} A_j\right) \\
&= \sum_{j=1}^{\infty} P(A_j) \\
&= \sum_{j=1}^n P(A_j) + \sum_{j=n+1}^{\infty} P(A_j) \\
&= \sum_{j=1}^n P(A_j) + \sum_{j=n+1}^{\infty} P(\emptyset) \\
&= \sum_{j=1}^n P(A_j) + \sum_{j=n+1}^{\infty} 0 \\
&= \sum_{j=1}^n P(A_j) + 0 \\
&= \sum_{j=1}^n P(A_j).
\end{aligned}$$

Corollary 2.1

□

2.3 Basic probability rules

We can deduce from the probability axioms several basic “rules” that will be fundamental going forward. To succeed in the course, you need to become fluent in the use of both the axioms and these basic rules.

2.3.1 Complement rule

Theorem 2.3. For any event $A \subseteq S$, $P(A) = 1 - P(A^c)$.

This will come in handy because quite often $P(A^c)$ is easier to compute than $P(A)$, and so the complement rule gives us permission to do the easier thing and then apply a simple formula to arrive at the probability we actually care about.

Proof. Let $A \subseteq S$. As we saw in Theorem 1.1, $A \cup A^c = S$, and since an element cannot be both in A and not in A at the same time, it is a disjoint union: $A \cap A^c = \emptyset$. Because $A \cup A^c$ is equal to S , their probabilities are also equal, and so we have

$$\begin{aligned} P(A \cup A^c) &= P(S) && \text{(Theorem 1.1: } A \cup A^c = S) \\ P(A \cup A^c) &= 1 && \text{(axiom: total measure one)} \\ P(A) + P(A^c) &= 1 && \text{(Theorem 2.2: finite additivity)} \\ P(A) &= 1 - P(A^c). \end{aligned}$$

Bada bing. □

Remark 2.6. As my first probability professor **Brian D. Jones** would always repeat back in the fall of 2014: “math is consistent!” The philosophers might say “them’s fightin’ words,” but to hell with them. BDJ is my guy. We know from Theorem 1.1 that $S^c = \emptyset$, and we know from the axiom of total measure 1 that $P(S) = 1$, so the complement rule says that

$$P(\emptyset) = P(S^c) = 1 - P(S) = 1 - 1 = 0,$$

which agrees with Theorem 2.1. If it didn’t, I’d be having a heart attack right now. But hey: math is consistent!

Remark 2.7. Having said all that, be careful! That sanity check we just did would *not* serve as a standalone proof that $P(\emptyset) = 0$. Why not? Because we used the complement rule, which was proven using finite additivity, which was proven using $P(\emptyset) = 0$:

$$\boxed{\text{Axioms}} \implies P(\emptyset) = 0 \implies \text{finite additivity} \implies \text{complement rule}.$$

You can’t use the complement rule (as we have proven it here) unless you *already know* that $P(\emptyset) = 0$. Therefore, using the complement rule to prove $P(\emptyset) = 0$ from scratch would be *circular*. As I said earlier, this course is all about the *downstream* consequences of the probability axioms. So let’s make sure the stream is flowing in one direction and not bending back on itself.

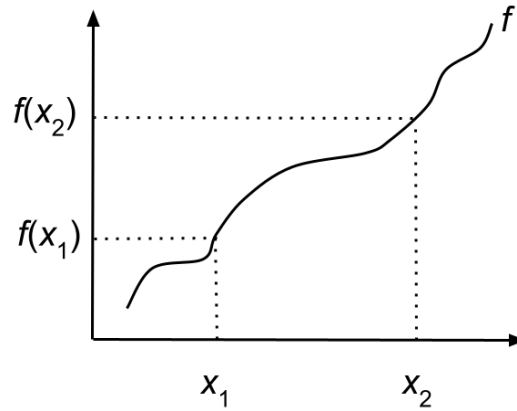


Figure 2.2: f is a monotone increasing function, so if the inputs are ordered $x_1 \leq x_2$, then the outputs have the same ordering: $f(x_1) \leq f(x_2)$. So we say f is an *order-preserving* map. The order of the original x -values was preserved after the function f was applied to them.

2.3.2 Monotonicity

Theorem 2.4. If $A \subseteq B$, then $P(A) \leq P(B)$.

“Monotonicity” means that the probability measure P is an order-preserving function on sets: if A is a “smaller” set than B in the sense that $A \subseteq B$, then correspondingly $P(A) \leq P(B)$. So the order is preserved after the P function is applied. This is similar to a monotone *increasing* function of real numbers like the one displayed in Figure 2.2.

Proof. To get the job done, we need to express B in terms of A . Recalling Theorem 1.5 and Figure 1.6, we know that $B = A \cup (B \cap A^c)$ and that this union is disjoint. As such:

$$\begin{aligned}
 P(A \cup (B \cap A^c)) &= P(B) \\
 P(A) + P(B \cap A^c) &= P(B) && \text{(Theorem 2.2: finite additivity)} \\
 P(A) &= P(B) - \underbrace{P(B \cap A^c)}_{\geq 0} \\
 P(A) &\leq P(B). && \text{(axiom: nonnegativity)}
 \end{aligned}$$

We showed that $P(A)$ was equal to $P(B)$ minus a quantity that is necessarily nonnegative by our second axiom. So if $P(B \cap A^c)$ is exactly zero, $P(A) = P(B)$. If it is positive, $P(A) < P(B)$. Those are the only options, so either way, we have the result. $\square$

We can now use this result for a cute sanity check:

Corollary 2.5. For any $A \subseteq S$, $0 \leq P(A) \leq 1$.

Proof. Let $A \subseteq S$. Our axioms tell us already that $0 \leq P(A)$ and $P(S) = 1$. Monotonicity gives us that $P(A) \leq P(S)$, and so the transitivity of $\leq$ gives that $0 \leq P(A) \leq 1$. $\square$

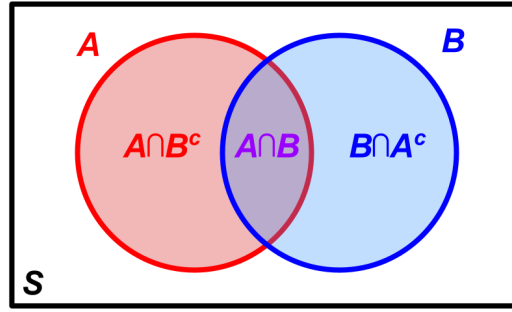


Figure 2.3

We surely already believed that probability was a number between zero and one, but this was not a property we explicitly imposed with an axiom. Instead, we assumed nonnegativity and total measure 1, and deduced this property as a consequence.

2.3.3 Law of inclusion/exclusion

Theorem 2.6. If $A, B \subseteq S$, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

This law is about the probability of a union of two events. We already have rules addressing this in the special case of two *disjoint* events, but not all events are disjoint, and so this rule completes the story by telling us what to do if there is overlap.

Proof. In all of our rules so far, the strategy has been to rewrite the set of interest as a disjoint union, apply countable additivity, and then do some algebra. So it shall be here. Inspecting Figure 2.3, we see that the sets A , B , and $A \cup B$ can all be rewritten as disjoint unions:

$$\begin{aligned} A &= (A \cap B) \cup (A \cap B^c) \\ B &= (A \cap B) \cup (B \cap A^c) \\ A \cup B &= (A \cap B^c) \cup (A \cap B) \cup (B \cap A^c). \end{aligned}$$

As such, we can apply finite additivity to each of these:

$$P(A) = P(A \cap B) + P(A \cap B^c) \tag{2.1}$$

$$P(B) = P(A \cap B) + P(B \cap A^c) \tag{2.2}$$

$$P(A \cup B) = P(A \cap B^c) + P(A \cap B) + P(B \cap A^c). \tag{2.3}$$

Rearranging (2.1) and (2.2) gives

$$\begin{aligned} P(A \cap B^c) &= P(A) - P(A \cap B) \\ P(B \cap A^c) &= P(B) - P(A \cap B), \end{aligned}$$

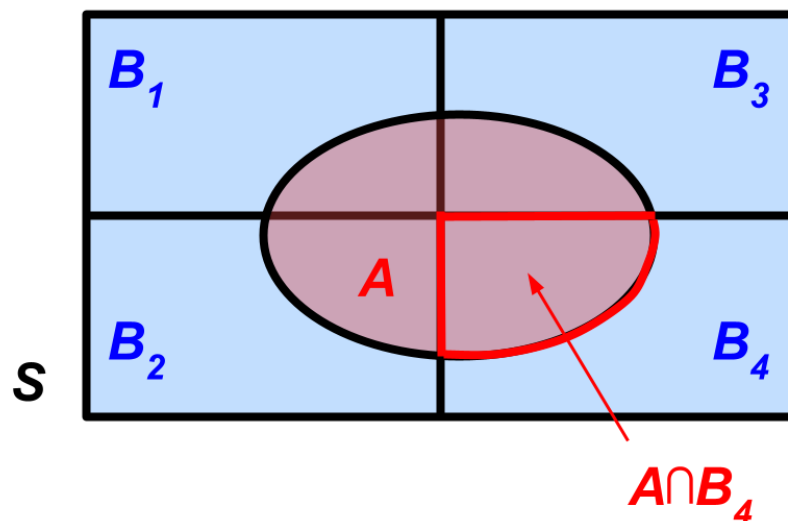


Figure 2.4

and plugging these expressions into (2.3) gives

$$\begin{aligned}
 P(A \cup B) &= P(A \cap B^c) + P(A \cap B) + P(B \cap A^c) \\
 &= P(A) - \cancel{P(A \cap B)} + \cancel{P(A \cap B)} + P(B) - P(A \cap B) \\
 &= P(A) + P(B) - P(A \cap B).
 \end{aligned}$$

□

The main idea of this proof is to avoid double counting the contribution of $A \cap B$ to the overall probability. The reason $P(A \cup B) = P(A) + P(B)$ does not remain the formula in the case of overlap is that $P(A)$ and $P(B)$ both “include” $P(A \cap B)$, and so the simple sum would incorrectly count it twice. To correct for this, we subtract it off once.

Example 2.1. In Hollywood, everyone is either rich or famous. 70% of people are rich and 50% are famous. How many are both? To answer this question, we use that $100\% = 70\% + 50\% - P(\text{both})$, so $P(\text{both}) = 20\%$.

2.3.4 Law of total probability

We start with a definition:

Definition 2.4. A collection of sets $B_1, B_2, B_3, \dots \subseteq S$ forms a **partition** of S if they cover everything ($\cup_{i=1}^{\infty} B_i = S$) and they do not overlap ($B_i \cap B_j = \emptyset$ for all distinct i and j).

Theorem 2.7. Let $B_1, B_2, B_3, \dots \subseteq S$ be a partition of S . Then for any $A \subseteq S$,

$$P(A) = \sum_{i=1}^{\infty} P(A \cap B_i).$$

An alternative name for this could be the “divide-and-conquer” law. It says that, in order to compute the “total” probability of A , we have the option to compute the separate probabilities of the parts of A that overlap with the components of the partition. Once we have these, we just add them up to get the overall $P(A)$. As with the complement rule, the reason we care about this is because, sometimes, computing the separate probabilities is easier. Furthermore, recalling Remark 2.4, Figure 2.4 displays a cartoon demonstrating that the law of total probability is perfectly analogous with a “law of total area” (or length, or volume, etc): if you partition the plane, the total area of a blob in the plane can be alternatively expressed as the sum of the smaller areas in each component of the partition. All we’re saying is that probability behaves exactly the same way.

Proof. Let $B_1, B_2, \dots \subseteq S$ partition S , and let $A \subseteq S$. We can rewrite A as a disjoint union:

$$\begin{aligned}
 A &= A \cap S \\
 &= A \cap \left(\bigcup_{i=1}^{\infty} B_i \right) \\
 &= A \cap (B_1 \cup B_2 \cup B_3 \cup \dots) \\
 &= (A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3) \cup \dots \quad (\text{distributive property}) \\
 &= \bigcup_{i=1}^{\infty} (A \cap B_i).
 \end{aligned}$$

Because the B_i are pairwise disjoint, so are the $A \cap B_i$ (show this!), and so the result follows from countable additivity:

$$P(A) = P\left(\bigcup_{i=1}^{\infty} (A \cap B_i)\right) = \sum_{i=1}^{\infty} P(A \cap B_i).$$

□

Remark 2.8. (“the move”) You just saw rigorous proofs of four fundamental probability rules: complement rule, monotonicity, the law of inclusion-exclusion, and the law of total probability. Please notice that in each case, we performed the same “move” somewhere in the proof: we used the set operations to rewrite a set as a disjoint union, and then we applied finite or countable additivity. Additivity is the tool that gives us permission to compute probabilities by...*adding*. It turns probability into an *arithmetic* problem, which is convenient, because some days it can feel like arithmetic is all I really know how to do. In all of the subsequent proofs, always keep an eye out for “the move.” It shows up a lot.

Remark 2.9. (do not do this on an exam!) Be careful that you don’t mix up operations on sets and operations on real numbers. $\cap$, $\cup$, and c are things you do to sets. $+$, $\times$, $-$, and $\div$ are things you do to real numbers like probabilities. So $P(A) \cup P(B)$ doesn’t mean anything. And $A + B$ doesn’t necessarily mean anything.

Remark 2.10. (“our cathedral”) We are investigating the downstream consequences of set theory and the probability axioms. Let’s map out everything we have proven so far, and what results we used to do it:

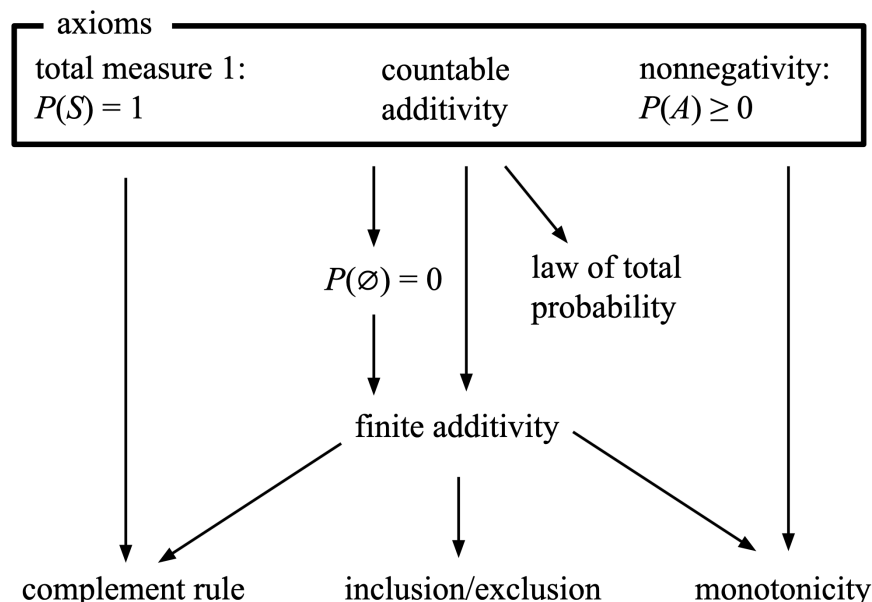


Figure 2.5: What did we use and when did we use it to prove the rules so far?

I hasten to make three points. Firstly, set theory is a prerequisite for *everything* in Figure 2.5. All along the way, we’re using those results too. Secondly, Figure 2.5 is not *unique*. This is how we happened to build *our* cathedral here in these notes, but there are other ways we might have developed the ideas. Thirdly, probability is not merely a list of rules that you memorize so that you can do word problems. It’s a structured chain of mathematical reasoning that we will continue to build on.

2.3.5 Advice for writing proofs

There is no generic algorithm or recipe for writing proofs. You just have to study a bunch of them, notice the patterns, and practice a lot yourself. But for the sorts of probability proofs that you will write in this class, I can offer some generic guidance to help you get started:

1. **Draw a picture.** Probability behaves like area, so see if you can make sense of the theorem statement by drawing and shading blobs;
2. **Take hints from the statement.** The statement of the theorem to be proven often gives a hint at what tools will be used. In “ $P(A) = 1 - P(A^c)$,” the presence of “1” tips you off that the first axiom will probably show up. In “ $P(A) \leq P(B)$,” the presence of an inequality tips you off that the second axiom will probably show up, and so on. In the future for example, if the theorem statement has probabilities *subtracted* somewhere, maybe that’s a clue that the complement rule or inclusion/exclusion will be used;
3. **Look out for “the move.”** I cannot guarantee that in 100% of all proofs, you will take a disjoint decomposition and apply additivity, but if that’s not literally what you do, you’re probably still using some other tool that was itself *enabled* by “the move” (eg. the complement rule). Anyway, “the move” is all about taking complicated sets and busting them into simpler disjoint pieces so that you can *add*. It’s always worth a try.